

ODESSKOYE, Russian Federation

WMO: 287970

Lat: 54.200N

Lon: 72.967E

Elev: 125

StdP: 99.84

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.1	-31.4	-36.8	0.1	-34.0	-34.0	0.2	-31.3	9.8	-11.1	8.3	-8.2	1.5	260	0.608

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	31.4	18.5	29.4	18.1	27.6	17.4	20.4	27.3	19.6	26.4	18.8	25.2	3.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.4	22.6	17.3	12.6	21.8	16.4	11.8	21.4	59.4	27.5	56.5	26.0	53.8	25.2	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.6	7.6	6.6	-35.9	34.2	3.5	1.7	-38.4	35.4	-40.4	36.4	-42.4	37.4	-45.0	38.6		
			-36.0	22.4	3.5	1.1	-38.5	23.2	-40.5	23.8	-42.5	24.4	-45.0	25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.2	-18.0	-16.0	-7.7	4.4	13.0	18.3	19.3	17.3	11.3	3.9	-6.8	-14.0
	DBStd	14.73	7.79	7.17	6.29	6.28	5.26	4.48	3.76	4.18	4.75	4.99	7.79	7.78
	HDD10.0	3851	867	728	550	186	30	2	0	1	40	197	505	745
	HDD18.3	6074	1126	962	808	417	178	56	34	71	218	447	755	1003
	CDD10.0	996	0	0	0	19	123	251	289	228	78	8	0	0
	CDD18.3	177	0	0	0	0	12	54	66	38	6	0	0	0
	CDH23.3	1906	0	0	0	6	200	598	626	396	79	1	0	0
	CDH26.7	597	0	0	0	1	51	202	194	128	21	0	0	0
Wind	WSAvg	3.1	3.0	3.1	3.5	3.5	3.4	2.8	2.5	2.5	2.8	3.1	3.3	3.4
Precipitation	PrecAvg	350	17	14	17	22	31	45	60	48	23	25	27	21
	PrecMax	483	41	25	39	66	77	90	176	95	58	63	71	41
	PrecMin	224	1	0	2	3	6	15	8	19	7	2	6	5
	PrecStd	75	9	6	11	15	16	19	40	23	13	15	16	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.3	0.7	6.4	23.8	31.2	33.1	33.9	33.4	29.3	19.8	7.7	1.2
		MCWB	-0.2	0.0	3.9	13.0	16.3	18.0	19.7	18.3	16.0	10.5	5.7	0.3
	2%	DB	-2.0	-1.1	3.1	20.1	27.8	31.2	30.6	30.1	25.6	17.1	5.8	0.0
		MCWB	-2.4	-1.7	1.6	10.9	14.8	18.1	19.2	18.4	14.7	9.8	4.3	-0.7
	5%	DB	-4.0	-2.9	1.8	17.6	25.5	29.2	28.7	27.4	22.1	14.2	3.9	-1.5
		MCWB	-4.4	-3.4	0.8	9.9	14.1	17.7	19.0	17.6	13.1	8.4	2.6	-2.1
	10%	DB	-6.2	-5.3	0.8	14.5	22.9	27.0	26.9	25.2	19.3	11.4	1.9	-3.8
		MCWB	-6.8	-5.8	-0.1	8.6	12.9	17.3	18.6	16.9	12.0	7.4	0.8	-4.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.2	0.1	3.9	13.9	17.6	21.0	21.7	20.7	17.7	12.7	6.3	0.6
		MCDB	0.3	0.5	6.4	21.6	27.0	28.8	27.6	27.1	27.2	18.2	7.2	0.9
	2%	WB	-2.5	-1.7	1.8	12.0	16.1	19.7	20.7	19.6	15.6	10.9	4.4	-0.6
		MCDB	-2.0	-1.1	2.8	18.8	25.3	27.7	27.3	26.4	22.7	15.3	5.6	-0.2
	5%	WB	-4.4	-3.4	0.9	10.4	14.9	18.7	20.0	18.7	14.1	9.1	2.5	-2.1
		MCDB	-4.0	-2.8	1.8	16.5	23.1	26.2	26.5	25.1	20.6	13.3	3.8	-1.5
	10%	WB	-6.8	-5.8	-0.1	8.9	13.8	17.8	19.2	17.7	12.7	7.5	0.9	-4.4
		MCDB	-6.3	-5.3	0.8	14.1	21.5	24.9	25.3	23.8	18.4	11.2	1.7	-3.8
Mean Daily Temperature Range	5% DB	MDBR	7.4	7.7	7.9	9.7	12.3	11.3	10.4	11.0	10.5	8.5	6.5	7.2
		MCDBR	8.3	7.4	6.5	14.9	15.6	14.4	13.3	14.6	14.9	13.3	6.8	7.3
		MCWBR	8.2	7.2	5.7	7.7	6.4	5.7	5.3	6.0	6.8	7.5	5.7	7.1
	5% WB	MCDBR	8.5	7.3	6.5	13.9	14.1	12.0	11.2	12.0	13.5	11.6	6.7	7.4
		MCWBR	8.4	7.2	5.9	7.7	6.9	5.5	5.2	5.5	6.7	7.2	5.8	7.3
Clear-Sky Solar Irradiance	taub	0.288	0.318	0.376	0.406	0.404	0.411	0.437	0.430	0.370	0.351	0.325	0.291	
	taud	2.046	2.017	1.945	2.142	2.221	2.288	2.203	2.190	2.315	2.254	2.117	2.019	
	Ebn at Noon	624	738	772	815	841	838	805	784	791	706	573	521	
	Edh at Noon	71	104	141	134	131	124	133	127	98	83	67	60	
All-Sky Solar Radiation	RadAvg	1.07	2.17	3.71	4.65	5.75	6.10	5.57	4.61	3.34	1.75	0.83	0.68	
	RadStd	0.08	0.08	0.33	0.40	0.49	0.32	0.42	0.34	0.24	0.23	0.11	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page